

Nina Quach

 q-nina

 q-nina.github.io

 s8niquac@uni-saarland.de

 nina-quach

 ORCID

Interests

Machine Learning – Human Computer Interaction – Data Science –
Eye-Tracking – Natural Language Processing – AI Safety

Education

- M.Sc.** Data Science and Artificial Intelligence, *Saarland University* 08/2023 to present
- **Master Thesis:** *Exploring the Impact of Quizzes in Elementary-Level Programming on Students' Problem-Solving Strategies* (in progress)
 - **Coursework:** Natural Language Processing, Reinforcement Learning, Machine Learning in Cybersecurity, Human Computer Interaction, Database Systems
- B.Sc.** Data Science and Artificial Intelligence, *Saarland University* 10/2019 to 08/2023
- **Bachelor Thesis:** *Classifying (un)known words in German based on reading behaviour; Self-Paced Reading and Eye-Tracking Studies* **Grade: 1.3**
 - Application Subject: Computational Linguistics
 - **Coursework:** Programming Foundations, Mathematical Foundations, Artificial Intelligence, Neural Networks, Data Science, Ethic for Nerds
- Abitur** Gymnasium Wendalinum (St. Wendel) 09/2011 to 06/2019

Experience

Learning to Adapt, Learning with Agency Lab – Saarland University

Student Research Assistant (HiWi)

Saarbrücken

- Supporting an interdisciplinary research project on goal setting and adaptive feedback.
- Performed descriptive and advanced statistical analyses including modeling and process mining, with a focus on data visualization for meaningful insights.
- Assisted in classroom studies with Gymnasium students.
- Contributing ontributing to the refinement of the AlgeSPACE platform.

08/2025 to 12/2025

ELO Digital Office GmbH

Working Student for Machine Learning

Saarbrücken

- Developing machine learning algorithms for specific real-world use cases.
- Analysing specifications or technical requirements to evaluate suitability of ML models.
- Fine-tuning LLMs for domain-specific tasks and performance optimisation.
- Designing Retrieval-Augmented Generation pipelines leveraging vector databases for efficient knowledge retrieval.
- Integrating machine learning and LLM-based solutions into existing workflows.

08/2023 to present

Publications

Ryzhova, M., Škrjanec, I., Quach, N., Chase, A. V., Ellsiepen, E., & Demberg, V. (2023, May). Word Familiarity Classification From a Single Trial Based on Eye-Movements. A Study in German and English. In *Proceedings of the 2023 Symposium on Eye Tracking Research and Applications*.

05/2023

[10.1145/3588015.3590118](https://doi.org/10.1145/3588015.3590118) 

Projects

Improving Programming Education with AI-Assisted Feedback

06/2024 – 09/2024

- as part of the seminar *Generative AI in Education* **Grade : 1.0**
- Designed a system to help intermediate-level Python students improve code efficiency and adhere to coding conventions beyond correctness.
- Implemented automated test input generation and feedback using domain-specific LLMs, comparison of zero-shot and one-shot hint generation
- Emphasized compact models for accessible, privacy-friendly deployment.

Analyzing Stroke Performance in High Performance Swimmer with Smart Paddles

06/2022 – 08/2022

- as part of *Project Seminar Data Science and Artificial Intelligence*
- Find quantitative criteria with data analysis and statistical methods to judge the swimming performance with data recorded by Smart Paddles.

Semi-Supervised Learning in Computer Vision

02/2022 – 03/2022

- as part of *Neural Networks: Theory and Implementation* **Grade : 1.4**
- Implemented existing semi-supervised learning algorithms. Designed and implemented own algorithm, which improved upon the existing algorithms.

Addressing OOV Words with Subword Representations

08/2021

- as part of *Statistical Natural Language Processing* **Grade : 1.0**
- Trained and improved language models for morphologically different languages by hyperparameter tuning and reducing OOV rates with subword segmentation.

Tutoring Experience

Supervising Teaching Assistant for Statistics Lab

04/2024 – 09/2024

Teaching Assistant for Statistics with R

10/2022 – 03/2023

Teaching Assistant for Statistics Lab

04/2022 – 09/2022

Teaching Assistant for Grundzüge von Algorithmen und Datenstrukturen

11/2021 – 03/2022

Teaching Assistant for Programming 1

11/2020 – 03/2021

Scholarships

Deutschlandstipendium

10/2019 – 03/2020

Skills

Programming Languages: Python, $\LaTeX$, R

Frameworks: PyTorch, Git

Software: Visual Studio, Pycharm, Azure AI Studio

Languages: German (Native), English (C1), Vietnamese (B1)